

VPA ICT — IGCSE Y9S2 Weeks 10–18: Data Manipulation & Presentations

Weeks 10 – 18: Data Manipulation & Presentations

Syllabus: CIE IGCSE ICT (0417) Topics 11–12 | Software: Excel/Calc + PowerPoint/Impress

Week 10 — Spreadsheet Setup & Data Entry

Syllabus: 11.1 Data Manipulation — Spreadsheets

Keywords: worksheet 工作表 (gōngzuò biǎo), workbook 工作簿 (gōngzuò bù), cell reference 单元格引用 (dānyuán gé yǐnyòng), absolute reference 绝对引用 (juéduì yǐnyòng), relative reference 相对引用 (xiāngduì yǐnyòng), named range 命名区域 (mìngmíng qūyù)

Activity 1 (15 min): Import CSV data into Excel. Check for formatting issues (dates, leading zeros).

Activity 2 (15 min): Absolute vs. relative reference — Create a multiplication table using $\$A\1 absolute reference.

Activity 3 (10 min): Named ranges — Name cell B2 as "TaxRate" and use $=B5*\text{TaxRate}$ in formula.

Exam Tip: Absolute references ($\$A\1) are essential for copying formulas. In exams, check if the question asks for "absolute cell reference" explicitly — missing the \$ signs loses marks.

Week 11 — Formulas & Functions

Syllabus: 11.2 Formulas and Functions

Keywords: formula 公式 (gōngshì), function 函数 (hánshù), SUM 求和 (qiúhé), AVERAGE 平均值 (píngjūn zhí), MAX 最大值 (zuìdà zhí), MIN 最小值 (zuìxiǎo zhí), COUNT 计数 (jìshù), COUNTIF 条件计数 (tiáojiàn jìshù), IF 条件函数 (tiáojiàn hánshù), VLOOKUP 垂直查找 (chuízhí cházhǎo)

Activity 1 (15 min): Gradebook functions — Use SUM, AVERAGE, MAX, MIN, COUNT for class scores.

Activity 2 (15 min): IF function — Grade calculator: $=\text{IF}(B2 \geq 80, "A", \text{IF}(B2 \geq 70, "B", \text{IF}(B2 \geq 60, "C", "D")))$

Activity 3 (10 min): VLOOKUP — Match student ID to name from a lookup table.

$=\text{VLOOKUP}(A2, \text{StudentTable}, 2, \text{FALSE})$

Week 12 — Conditional Formatting & Data Validation

Syllabus: 11.3 Formatting and Validation

Activity 1 (15 min): Apply conditional formatting: highlight scores > 90 in green.

Activity 2 (15 min): Data validation — Create dropdown list for "Subject" column. Add error message for invalid entry.

Activity 3 (10 min): Protection — Lock all cells except input cells. Test by trying to edit.

Week 13 — Charts & Graphs

Syllabus: 11.4 Charts

Keywords: chart wizard 图表向导 (túbiǎo xiàngdǎo), axis 轴 (zhóu), legend 图例 (tú lì), data label 数据标签 (shùjù biāoqiān), trend line 趋势线 (qūshì xiàn), secondary axis 次坐标轴 (cì zuòbiāo zhóu)

Activity 1 (15 min): Create bar chart from sales data. Add title, axis labels, data labels.

Activity 2 (15 min): Combo chart — Bar for sales, line for profit on secondary axis.

Activity 3 (10 min): Chart formatting — Change colors, add trend line, modify legend position.

Week 14 — Pivot Tables & Sorting

Syllabus: 11.5 Pivot Tables

Keywords: pivot table 数据透视表 (shùjù tòushì biǎo), row field 行字段 (háng zìduàn), column field 列字段 (liè zìduàn), data field 数据字段 (shùjù zìduàn), filter field 筛选字段 (shāixuǎn zìduàn)

Activity 1 (20 min): Create pivot table: summarize sales by region and product category.

Activity 2 (15 min): Add filter — Show only Q1 and Q2 data.

Activity 3 (5 min): "When is a pivot table better than a regular table?" Discussion.

Exam Tip: Pivot tables frequently appear in Paper 2. Remember: drag fields to correct areas (Row, Column, Data, Filter). The data field usually needs SUM or AVERAGE function.

Week 15 — Presentation Basics

Syllabus: 12.1 Presentation Software

Keywords: slide 幻灯片 (huàndēngpiàn), master slide 母版幻灯片 (mǔbǎn huàndēngpiàn), layout 版式 (bǎnshì), placeholder 占位符 (zhàn wèi fú), slide show 幻灯片放映 (huàndēngpiàn fàngyǐng), speaker notes 演讲者备注 (yǎnjiǎng zhě bèizhù)

Activity 1 (15 min): Master slide setup — Create consistent title, footer, and logo on master slide.

Activity 2 (15 min): Apply layouts — Title slide, Title and Content, Two Content, Comparison.

Activity 3 (10 min): Slide size setup — 4:3 vs. 16:9. When to use each?

Week 16 — Animations, Transitions & Multimedia

Syllabus: 12.2–12.3 Animations and Multimedia

Keywords: animation 动画 (dòng huà), transition 切换 (qiē huàn), entrance 进入 (jìn rù), emphasis 强调 (qiáng diào), exit 退出 (tuì chū), motion path 动作路径 (dòng zuò lù jìng), trigger 触发器 (chù fā qì)

Activity 1 (15 min): Add entrance animations to bullet points. Set to appear "On Click."

Activity 2 (15 min): Slide transitions — Apply subtle transition to all slides. Set automatic timing (5 seconds).

Activity 3 (10 min): Insert video and audio. Set to play automatically. Trim video to first 30 seconds.

Exam Tip: For "automatic slide show" questions, you must set BOTH: 1) Slide transition timing (e.g., 5 seconds), AND 2) "Start After Previous" for animations. Missing either loses marks.

Week 17 — Advanced Presentation Features

Syllabus: 12.4 Advanced Features

Activity 1 (15 min): Hyperlinks and action buttons — Create interactive quiz: click answer → jump to correct/incorrect slide.

Activity 2 (15 min): Custom slide show — Create two versions: full version (10 slides) and summary version (5 slides).

Activity 3 (10 min): Export to PDF and video. Compare file sizes and use cases.

Week 18 — End of Semester Assessment

Mock Paper 2 Section B (40 min):

Open provided spreadsheet with raw data

Add 3 calculated columns using formulas

Create 2 charts (bar and pie) with titles and labels

Create a pivot table summarizing data by category

Apply conditional formatting to highlight top 10%

Presentation Task (40 min):

Create 8-slide presentation from given outline

Apply master slide with logo and footer

Insert 2 images and 1 chart from spreadsheet

Add entrance animations and slide transitions

Set up as automatic slide show (5 seconds per slide)

Paper 2 Marking Checklist

Data Manipulation (40 marks typical):

- ☐ Formulas correctly entered and copied
- ☐ Absolute/relative references used appropriately
- ☐ Functions (SUM, AVERAGE, IF, VLOOKUP, COUNTIF) correct
- ☐ Conditional formatting applied correctly
- ☐ Data validation with appropriate criteria
- ☐ Charts have correct type, title, labeled axes
- ☐ Pivot table configured correctly
- ☐ Data sorted and/or filtered as requested

Presentations (20 marks typical):

- ☐ Master slide applied consistently
- ☐ Correct layouts used for each slide
- ☐ Animations and transitions applied
- ☐ Multimedia inserted and configured
- ☐ Automatic timings set correctly